Erik Marsja | Curriculum Vitae

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Positions

Jan 2023 - **Assistant Professor**

Disability Research Division, Department of Behavioural Sciences and Learning (IBL), Linköping University.

Jan 2022 -

Dec 2022 **Researcher**

People in the transport system, Swedish National Road and Transport Research Institute, Linköping, Sweden. Working 80% of a full-time position.

Jan 2022 -

Nov 2021 **Postdoctoral Researcher**

Disability Research Division, Department of Behavioural Sciences and Learning (IBL), Linköping University.

Analyzing data from a large database (i.e., n200) and preparing manuscripts in the field of cognitive hearing science. Conducting qualitative research and about digitalization, listening effort, and work-life (see Funding and Grants). Working 20% of a full-time position.

Jan 2019 -

Dec 2021 **Postdoctoral Researcher**

Disability Research Division, Department of Behavioural Sciences and Learning (IBL), Linköping University.

Analyzing data from a large database (i.e., n200) and preparing manuscripts in the field of cognitive hearing science. Responsible for arranging the monthlyHEAD/DRD seminar series (august 2020 – august 2021). This includes contacting national and international researchers, booking flights and hotels, and setting up equipment for streaming the seminars online. Teaching and supervising.

Aug 2018 -

Oct 2018 **Postdoctoral Research Assistant**

Department of Psychology, Umeå University.

Project: How does humans perception of safety differ in a broken communication chain compared to an unbroken communication chain?

Literature study commissioned by SOS Alarm, an emergency agency (e.g., public-safety answering point), with the results disseminating in a report.

Oct 2012 - June 2018 **PhD Student and Teaching Assistant**

Department of Psychology, Umeå University.

Planning of studies, programming of experiments, literature search, data analysis, scientific communication, and many more. See the section "Teaching responsibilities" for an overview of my pedagogical experience.

June 2011 **Undergrad. Research Assistant**

Department of Psychology, Umeå University.

Recruitment of participants and data collection, commenting on later manuscript that ended up in a published paper (see publication 2014).

Degrees and Titles

2026 **Docent (Associate Professor) in Disability Research**

Department of Behavioural Sciences and Learning, Linköping University.

2017 **Ph.D. in Psychology**

Department of Psychology, Umeå University.

Thesis title: Attention capture by sudden and unexpected changes: a multisensory perspective.

Available from the DiVA Database: http://umu.diva-portal.org/smash/record.jsf?pid=diva2

Supervisors: Associate Professor Jessica K. Ljungberg, Professor Gregory Neely, & Dr. Patrik Hansson

2012 **M.Sc. in Cognitive Science**

Department of Psychology, Umeå University.

Thesis title: Attention Capture: The Impact of Change in Spatial Sound Source on Behavior.

Supervisor: Associate Professor Jessica K. Ljungberg

2011 **B.Sc. in Cognitive Science**

Department of Psychology, Umeå University.

Thesis title: Attention Capture: Studying the Distracting Effect of One's Own Name.

Available from DiVA Database: http://urn.kb.se/resolve?urn=urn:nbn:se:umu:diva-46607.

Supervisor: Associate Professor Jessica K. Ljungberg

Publications

Accepted/Published

19 peer-reviewed journal articles (7 as first author; 1 as last author).

- Ning, R., **Marsja**, E., & Holmer, E. (Accepted). The association between fluid intelligence and speech-in-noise performance in hearing-aid users is moderated by masker type and intelligibility criterion. *Journal of Speech, Language, and Hearing Research*.
- **Marsja**, E., Rodrigues, V., Mortazavi, A., & & Blomkvist, J. (2027). Public transport for people with hearing impairment: A scoping review. *Journal of Transport & Health, 52*, 102454. `https://doi.org/10.1016/j.jth.2026.102454`
- **Marsja**, E., Signoret, C., & Stenbäck, V. (2026). The digital workplace and meeting accessibility: A qualitative study on listening effort in video meetings for employees with hearing loss. *WORK: A Journal of Prevention, Assessment & Rehabilitation, 84*(1), 197-209. `https://doi.org/10.1177/10519815251398498` (PDF)

- Ning, R., **Marsja**, E., & Holmer, E. (2026). Poorer Sentence Reading Comprehension With Increased Reliance on Working Memory in Adults With Hearing Loss. *Journal of Speech, Language, and Hearing Research, 69*(2), 807–824. https://doi.org/10.1044/2025_jslhr-25-00187 (PDF)
- Rönnberg, J., **Marsja**, E., Danielsson, H., Holmer, E., & Dahlström, Ö. (2026). Beyond Accuracy: Cognitive Speed in Speech-in-Noise Processing and Its Implications for the Ease of Language Understanding Model. *Journal of Speech, Language, and Hearing Research, 69*(5), 2303–2322. https://doi.org/10.1044/2026_JSLHR-25-00773 (PDF)
- Thorslund, B., Söderman, J., **Marsja**, E., & Selander, H. (2026). Validating variations of situations in a driving simulator screening test. *Transportation Research Part F: Traffic Psychology and Behaviour, 119*, 103608. https://doi.org/10.1016/j.trf.2026.103608 (PDF)
- **Marsja**, E., Holmer, E., Stenbäck, V., Micula, A., Tirado, C., Danielsson, H., & Rönnberg, J. (2025). Fluid Intelligence Partially Mediates the Effect of Working Memory on Speech Recognition in Noise. *Journal of Speech, Language, and Hearing Research, 68*(1), 399–410. https://doi.org/10.1044/2024_JSLHR-24-00465 (PDF)
- **Marsja**, E., Thellman, S., & Anund, A. (2025). Trust in Automated Shuttle Buses Predicts Intentions to Use and Behavioral Expectations. *Transportation Research Interdisciplinary Perspectives, 33*, 101601. https://doi.org/10.1016/j.trip.2025.101601 (PDF)
- Stenbäck, V., Hällgren, M., & **Marsja**, E. (2025). Speech Recognition Under Masking: Age, Hearing, and Machine Learning Classification. *Acta Psychologica, 259*, 105461. https://doi.org/10.1016/j.actpsy.2025.105461 (PDF)
- Marsh, J. E., Vachon, F., Sörqvist, P., **Marsja**, E., Röer, J. P., Richardson, B. H., & Ljungberg, K. J. (2024). Irrelevant Changing-State Vibrotactile Stimuli Disrupt Verbal Serial Recall: Implications for Theories of Interference in Short-Term Memory. *Journal of Cognitive Psychology, 36*(1), 78–100. https://doi.org/10.1080/20445911.2023.2198065 (PDF)
- Elbe, P., **Marsja**, E., Sörman, D., Vega-Mendoza, M., Nyberg, L., & Ljungberg, K. J. (2024). Effects of Auditory and Tactile Distraction in Adults With Low and High ADHD Symptoms. *Journal of Cognitive Psychology, 36*(5), 645–656. https://doi.org/10.1080/20445911.2024.2349331 (PDF)
- Stenbäck, V., **Marsja**, E., Ellis, R., & Rönnberg, J. (2023). Relationships Between Behavioural and Self-Report Measures in Speech Recognition in Noise. *International Journal of Audiology, 62*(2), 101–109. https://doi.org/10.1080/14992027.2022.2047232 (PDF)
- **Marsja**, E., Stenbäck, V., Moradi, S., Danielsson, H., & Rönnberg, J. (2022). Is Having Hearing Loss Fundamentally Different? Multi-Group Structural Equation Modeling of the Effect of Cognitive Functioning on Speech Identification. *Ear and Hearing, 43*(5), 1437–1446. https://doi.org/10.1097/AUD.0000000000001196 (PDF)
- Stenbäck, V., **Marsja**, E., Hällgren, M., Lyxell, B., & Larsby, B. (2022). Informational Masking and Listening Effort in Speech-Recognition-in-Noise – The Role of Working Memory Capacity and Inhibitory Control in Older Adults With and Without Hearing Impairment. *Journal of Speech, Language, and Hearing Research, 65*(11), 4417–4428. https://doi.org/10.1044/2022_JSLHR-21-00674 (PDF)
- Stenbäck, V., **Marsja**, E., Hällgren, M., Lyxell, B., & Larsby, B. (2021). The Contribution of Age, Working Memory Capacity, and Inhibitory Control on Speech Recognition in Noise in Young and Older Adult Listeners. *Journal of Speech, Language, and Hearing Research, 64*(11), 4513–4523. https://doi.org/10.1044/2021_JSLHR-20-00251

- Rosa, E., **Marsja**, E., & Ljungberg, J. K. (2020). Exploring Residual Capacity: The Effectiveness of a Vibrotactile Warning During Increasing Levels of Mental Workload in Simulated Flight Tasks. *Aviation Psychology and Applied Human Factors, 10*(1), 13–23. https://doi.org/10.1027/2192-0923/a000180
- **Marsja**, E., Marsh, J. E., Hansson, P., & Neely, G. (2019). Examining the Role of Spatial Changes in Bimodal and Uni-Modal To-Be-Ignored Stimuli and How They Affect Short-Term Memory Processes. *Frontiers in Psychology, 10*. https://doi.org/10.3389/fpsyg.2019.00299 (PDF)
- **Marsja**, E., Neely, G., & Ljungberg, K. J. (2018). Investigating Deviance Distraction and the Impact of the Modality of the To-Be-Ignored Stimuli. *Experimental Psychology, 65*(2), 61–70. https://doi.org/10.1027/1618-3169/a000390 (PDF)
- Ljungberg, K. J., Parmentier, F. B. R., Jones, D. M., **Marsja**, E., & Neely, G. (2014). "What's in a Name?" "No More Than When It's Mine Own." Evidence From Auditory Oddball Distraction. *Acta Psychologica, 150C*, 161–166. https://doi.org/10.1016/j.actpsy.2014.05.009

Ongoing Manuscripts and Registered Reports

9 ongoing manuscripts: 1 Stage 1 Registered Report with in-principle acceptance, 1 under revision, 7 submitted, and 0 in preparation.

2 manuscripts currently undergoing double-blind peer review are omitted from the public list below to preserve author anonymity.

- De Temmerman, K., Topor, M., **Marsja**, E., Martin, L., & Signoret, C. (Stage 1 Registered Report with in-principle acceptance). Mapping Auditory Neuroplasticity: A Registered Report Systematic Review of Speech Perception Interventions. https://rr.peercommunityin.org/articles/rec?id=2735.
- Signoret, C., Rönnberg, J., & **Marsja**, E. (Submitted). High-Frequency Hearing Loss Is the Strongest Predictor of Speech Perception in Noise for Hearing-Aid Users.
- Tonneman, T., Stenbäck, V., Ingo, E., & **Marsja**, E. (Submitted). From Accessibility Needs to Platform Gaps: A Criterion-Based Assessment of Videoconferencing for Employees with Hearing Loss.
- Larsson, S., Danielsson, H., **Marsja**, E., & Sjörs Dahlman, A. (Submitted). Feasibility of Using Virtual Reality When Investigating Street-Crossing Behavior in Older Pedestrians.
- **Marsja**, E., & Andin, J. (Submitted). Hearing Status and Hearing-Aid Use in Relation to Long-Term Cognitive Health Trajectories: A Population-Based Longitudinal Study.
- **Marsja**, E., Pernodd, J., Rodrigues, V., Mortazavi, A., & Blomkvist, J. (Submitted). Information Access for Older Passengers: A Mixed-Methods Study of Bus Travel With and Without Hearing Loss.
- **Marsja**, E., Luccini, F., Stenbäck, V., Ingo, E., Danielsson, H., Dahlström, Ö., & Holmer, E. (Submitted). Internal Structure and Measurement Invariance of the Swedish 12-Item Speech, Spatial, and Qualities of Hearing Scale.

Conference Presentations

5 oral presentations and 16 posters.

- **Marsja**, E., Stenbäck, V., Danielsson, H., & Ingo, E. (2026, June). Effortful Listening in Digital Work Environments: A Qualitative Study of Participation and Communication Demands Among

Employees With Hearing Loss. Hearing Across the Lifespan (HEAL), Padova, Italy. **Oral presentation**.

- **Marsja**, E., Granlund, M., Brandén, M., Olsson, L., & Danielsson, H. (2025, June). The Impact of Personal Assistance on Family Disposable Income: Insights from Swedish Register Data. 13th European Conference on Psychological Theory and Research on Intellectual and Developmental Disabilities (ECIDD), Gothenburg, Sweden. **Oral presentation**.
- **Marsja**, E., Ning, R., & Holmer, E. (2025, January). The role of noise type and task demand in moderating the effect of hearing aid signal processing for speech in noise perception. Speech in Noise Workshop, Lancaster, United Kingdom. **Poster**.
- Ning, R., **Marsja**, E., & Holmer, E. (2025, January). Testing the moderating effect of interference control on the impact of fluid intelligence on SPiN. Speech in Noise Workshop, Lancaster, United Kingdom. **Poster**.
- Holmer, E., Ning, R., & **Marsja**, E. (2025, January). The strength of association between cognitive capacity and effective speech recognition in noise is moderated by noise and task conditions. Speech in Noise Workshop, Lancaster, United Kingdom. **Poster**.
- Thurén, F., Wadenholt, G., & **Marsja**, E. (2024, October). Towards a Self-Driving Future: A study on the factors influencing behavioral intentions to use autonomous buses. 19th SweCog Conference. **Poster** (Short paper; pp. 135–137).
- **Marsja**, E., Holmer, E., & Danielsson, H. (2024, January). Interplay between working memory and speech recognition declines over time. Speech in Noise Workshop, Potsdam, Germany. **Poster**.
- Danielsson, H., & **Marsja**, E. (2024, January). Speech in noise in the n200 study in Linköping Sweden. Speech in Noise Workshop, Potsdam, Germany. **Oral presentation** (Danielsson presented).
- **Marsja**, E., Lucchini, F., Stenbäck, V., Holmer, E., & Danielsson, H. (2023, August). Exploring the Validity of the Speech, Spatial, and Qualities of Hearing Questionnaire in Swedish: A Psychometric Analysis. International Symposium on Auditory and Audiological Research, Nyborg Strand, Denmark. **Poster**.
- Thellman, S., **Marsja**, E., Anund, A., & Ziemke, T. (2023). Will It Yield: Expectations on Automated Shuttle Bus Interactions With Pedestrians and Bicyclists. Companion of the 2023 ACM/IEEE International Conference on Human-Robot Interaction. **Late-breaking report** (Short paper; pp. 292–296).
- **Marsja**, E., Danielsson, H., Stenbäck, V., Moradi, S., & Rönnberg, J. (2019, November). Examining how Cognitive Functioning, Aging, and Hearing Loss, Affect Speech-in-Noise Performance. Aging and Speech Communication Conference, Tampa, Florida, USA. **Poster**.
- Stenbäck, V., **Marsja**, E., Danielsson, H., & Rönnberg, J. (2019, November). Logical and Auditory Inference Making: Performance in the HINT in normally-hearing and hearing-impaired listeners. Aging and Speech Communication Conference, Tampa, Florida, USA. **Poster**.
- Bampouni, E., **Marsja**, E., Sörman, D. E., & Ljungberg, K. J. (2019, November). Do Action Gamers Have Enhanced Visual Search Skills? A Realistic Task Approach. 15th SweCog Conference of the Swedish Cognitive Science Society, Umeå, Sweden. **Poster**.
- **Marsja**, E., Danielsson, H., Stenbäck, V., Moradi, S., & Rönnberg, J. (2019, June). Examining Relationship Amongst Cognition, Hearing Loss, Age, & Speech in Noise. Cognitive Hearing Science for Communication, Linköping, Sweden. **Poster**.

- Stenbäck, V., Moradi, S., **Marsja**, E., Danielsson, H., & Rönnberg, J. (2019, June). Logical and auditory inference-making in normally-hearing and hearing-impaired listeners. Cognitive Hearing Science for Communication, Linköping, Sweden. **Poster**.
- **Marsja**, E., Marsh, J. E., Neely, G., Hansson, P., & Ljungberg, K. J. (2017, April). Domain-generality or domain-specificity of short-term memory: Insights from a multisensory distraction paradigm. Re-thinking the Senses Spring School, Dubrovnik, Croatia. **Poster**.
- **Marsja**, E., Marsh, J. E., Neely, G., Hansson, P., & Ljungberg, K. J. (2016, September). Do Spatial Changes in Sounds and Vibrations Affect Visuo-spatial and Verbal Short-Term Memory?. Attention and Control: Insights from Distraction Workshop, University of Central Lancashire, Preston, United Kingdom. **Invited presentation**.
- **Marsja**, E., Marsh, J. E., Neely, G., Hansson, P., & Ljungberg, K. J. (2016, June). Spatial Change in Multisensory Distractors Impact on Verbal and Spatial Short-Term Memory. 17th Annual Meeting of the International Multisensory Research Forum, Suzhou, China. **Oral presentation**.
- **Marsja**, E., Neely, G., Ma, L., & Ljungberg, K. J. (2015, August). Cross-modality matches of intensity and attention capture dimensions of auditory and vibrotactile stimuli. Fechner Day 2015: 31st Annual Meeting of the International Society for Psychophysics, Québec, Canada. **Poster**.
- **Marsja**, E., Neely, G., Parmentier, F. B. R., & Ljungberg, K. J. (2014, October). Deviance Distraction Is Contingent on Stimuli Being Presented within the Same Modality. Psychonomic Society's 55th Annual Meeting, Long Beach, California, USA. **Poster**.
- Ljungberg, K. J., Parmentier, F. B. R., **Marsja**, E., Neely, G., & Jones, D. (2014, January). Any Tom, Dick, or Harry will do: Hearing one's own name distracts no more than any other in a cross-modal oddball task. Experimental Psychology Society Meeting, London, United Kingdom. **Poster**.
- **Marsja**, E., Neely, G., Parmentier, F. B. R., & Ljungberg, K. J. (2013, October). Maintenance of the distractive effect of deviating vibrotactile stimuli in a cross-modal oddball paradigm. 29th Annual Meeting of the International Society of Psychophysics, Freiburg, Germany. **Poster**.

Popular Science and Education-Related Articles

Marsja, E. (2016, July). Python Programming in Psychology – From Data Collection to Analysis. *JEPS Bulletin - The Official Blog of the Journal of European Psychology Students*. **Invited Blog Post**. Retrieved from `http://blog.efpsa.org/2016/07/12/python-programming-in-psychology-from-data-collection-to-analysis/`

Science Outreach

Marsja, E. (October, 2024). Listening Effort in Video Meetings: Positive and Negative Experiences of Employees with Hearing Loss. Presentation for Employed (Professional) Members of the Swedish Association of the Hard of Hearing People, Linköping, Sweden.

Marsja, E. (November, 2022). The Contribution of Age, Hearing Status, Cognition, Background Masker on Speech Recognition in Noise and Listening Effort. Presentation at the Journal Club at Oticon Medical, Nice, France.

Marsja, E., & Stenbäck, V., (June, 2022). Current and Ongoing Projects on Hearing loss, Cognition, Psychosocial Perspectives, Video Meetings and Listening Effort. Presentation for the Hearing Clinics (Öron- näs- och halsklinikerna), Region Östergötland, Sweden.

Research Grants and Funding

Funding as project leader: 3 860 739 SEK. **Total funding:** 8 030 739 SEK.

- **103 600 SEK** from Riksbankens Jubileumsfond (RJ Research Initiation) for hosting the workshop *Cognitive Hearing Science Across Systems: From Mechanisms to Societal Outcomes*, 2026.
- **3 391 904 SEK** from Afa Försäkring for the project *Digital Arbetsmiljö för Alla – Riktlinjer och Policyrekommendationer för Delaktighet vid Hörselnedsättning (Digital Work Environment for All – Guidelines and Policy Recommendations for Participation for People with Hearing Loss)*, 2025.
- **4 170 000 SEK** from Swedish Research Council for the project *The temporal architecture of imitative interaction during the first three months of life* (PI: Prof. Emeritus Heimann), 2023.
- **299 235 SEK** from Hörselforskningsfonden (the Hearing Research Fund) for the project *Lyssningsansträngning hos individer med hörselnedsättning på grund av ökad digitalisering i arbetslivet (Listening effort in individuals with hearing impairment due to increased digitalisation in working life)*, 2021.
- **6 000 SEK** from Department of Psychology, Umeå University for participation in the Rethinking the Senses Spring School, Dubrovnik, Croatia, 2017.
- **8 000 SEK** from Department of Psychology, Umeå University for participation in the 17th International Multisensory Research Forum, 15–18 June, Suzhou, China, 2016.
- **12 000 SEK** from Faculty of Social Sciences, Umeå University for participation in the workshop Attention and Control: Insights from Distraction, and a research visit at the University of Central Lancashire, Preston, United Kingdom, 2016.
- **15 000 SEK** from Lars Hiertas Minnesfond for the project *Är korttidsminnet domän-generellt eller domän-specifikt? (Is short-term memory domain-general or domain specific?)*, 2016.
- **9 000 SEK** from Knut och Alice Wallenbergs Stiftelse for participation in the Psychonomic Society's 55th Annual Meeting, 20–23 November, and APCAM, 20 November, Long Beach, USA, 2014.
- **10 000 SEK** from JC Kempes minnesfond for the project *Is everyday distractibility related to attention capture by vibrating deviants?*, 2014.
- **6 000 SEK** from Department of Psychology, Umeå University for participation in Fechner Day 2013 (the 29th Annual Meeting of the International Society for Psychophysics), 21–25 October, Freiburg im Breisgau, Germany, 2013.

Peer Review

26 peer reviews completed for the following journals:

Acta Psychologica; American Journal of Audiology; Attention, Perception, & Psychophysics; Auditory Perception & Cognition; BMC Psychology; BMC Public Health; Brain and Cognition; Brain and Language; Experimental Gerontology; Frontiers In Future Transportation; Humanities & Social Sciences Communications; International Journal of Clinical Practice; Journal of Cognitive Psychology; Journal of Speech, Language, and Hearing Research; Journal of the International Neuropsychological Society; Memory & Cognition; NeuroImage; Science Progress; Scientific Reports; The Journal of the Acoustical Society of America; Transportation Research Part F: Traffic Psychology and Behaviour; Trends in Hearing; Work.

Skills & Training

Training

2017 Spring School *Re-Thinking the Senses*, Inter-University Centre, Dubrovnik, Croatia

Teaching

Primarily in:

- Research Methods & Statistics
- Cognitive Psychology (attention, perception, & working memory)

All lectures, seminars, lab demonstrations, supervision of group projects (both involving empirical and applied projects), and supervision of thesis' have been given at the Department of Psychology, Umeå University, Sweden (2014 - 2017) and the Department of Behavioral Science and Learning, Linköping University (2019 -). At Linköping University, I have also held workshops and supervised projects. I have taught 2490 and 950 clock hours at Linköping and Umeå University, respectively. See Table 1 for an overview of my teaching responsibilities, clock hours, and courses.

Supervision of Bachelor Students (Undergraduate Level)

Pernodd, J. (2024). Department of Computer and Information Science, Linköping University. "Påverkan av Hörselnedsättning på Äldres Ögonrörelsemönster och Upplevelse i Kollektivtrafiken (The Effect of Hearing Impairment on Eye Movement Patterns and Experience in Public Transportation for the Elderly)". Bachelor's Thesis in Cognitive Science (18 ECTS). *Main supervisor*

Pusker, M. (2024). Department of Computer and Information Science, Linköping University. "Lyssna på Tystnaden: En studie om upplevelserna av att resa med autonom kollektivtrafik för individer med hörselnedsättning (Listen to the Silence : A Study on the Experiences of Traveling with Autonomous Public Transport for Individuals with Hearing Impairment)". Bachelor's Thesis in Cognitive Science (18 ECTS). *Main supervisor*

Turén, F. (2024). Department of Computer and Information Science, Linköping University. "Mot en Självkörande Framtid: En studie i vilka faktorer som påverkar beteendemässiga intentioner att användasjälvkörande bussar (Towards a Self-Driving Future : A study on the factors influencing behavioral intentions to use autonomous buses)". Bachelor's Thesis in Cognitive Science (18 ECTS). *Main supervisor*

Fernlund, F. (2023). Department of Computer and Information Science, Linköping University. "Examining the effects of text support and noise during video meetings on listening effort and comprehension". Bachelor's Thesis in Cognitive Science (18 ECTS). *Main supervisor*

Lundgren, E. (2023). Department of Computer and Information Science, Linköping University. "Delays in Videoconferencing - The Effects of Asynchronous Speech on Listening Effort and Comprehension". Bachelor's Thesis in Cognitive Science (18 ECTS). *Main supervisor*

Bernsland, A. (2023). Department of Computer and Information Science, Linköping University. " 'Disgusting pictures' - A study exploring webpages´ experienced credibility based upon their visual design elements." Bachelor's Thesis in Cognitive Science (18 ECTS). *Main supervisor*

Jerdhaf, O. (2021). Department of Computer and Information Science, Linköping University. "Discovering Implant Terms in Medical Records". Bachelor's Thesis in Cognitive Science (18 ECTS). *Main supervisor*

Bridal, O. (2021). Department of Computer and Information Science, Linköping University. "Named-entity recognition with BERT for anonymization of medical records". Bachelor's Thesis in Cognitive Science (18 ECTS). *Main supervisor*

Mattila, M. (2021). Department of Computer and Information Science, Linköping University. "Synthetic Image Generation Using GANs – Generating Class-Specific Images of Bacterial Growth". Bachelor's Thesis in Cognitive Science (18 ECTS). *Main supervisor*

Rombo, A. (2020). Department of Computer and Information Science, Linköping University. "Self-determination perceived by users in support services pursuant to LSS - An analysis on a municipal level. Bachelor's Thesis in Cognitive Science (18 ECTS). *Main supervisor*

Lindberg, F. (2020). Department of Computer and Information Science, Linköping University. "Hur ungas attityder kring hörselnedsättningar orsakade av fritidsbuller påverkas av deras koppling till sitt framtida jag."(How young people's attitudes to hearing impairments caused by leisure noise are affected by their connection to their future selves.) Bachelor's Thesis in Cognitive Science (18 ECTS). *Main supervisor*

Dahlgren, S. (2020) Department of Computer and Information Science, Linköping University. "The association between cognition and speech-in-noise perception - Investigating the link between speech-in-noise perception and fluid intelligence in people with and without hearing loss". Bachelor's Thesis in Cognitive Science (18 ECTS). *Main supervisor*

Supervision of Master Students (Graduate Level)

Borgstrand, M. & Malm, L. (2024). Department of Behavioural Sciences and Learning, Linköping Unviersity. "Jag behövde bara ett annat sätt att förstå: En intervjustudie om unga kvinnor med ADHD och deras upplevelser av matematikundervisningen i grundskolan (I Just Needed Another Way to Understand : An Interview Study about Young Women with ADHD and Their Experiences of Mathematics Education in Primary School)". Master's Thesis in Special Needs (15 ECTS). *Main supervisor*

Andersson, J. & Sjöström, G. (2024). Department of Behavioural Sciences and Learning, Linköping Unviersity. "Visuellt brusreducerande glasögon - kan det hjälpa elever med intellektuell funktionsnedsättning att rikta och bibehålla fokus/uppmärksamhet? (Can QuietFrames Glasses Help Students with Intellectual Disabilities to Retain Focus/Attention?)". Master's Thesis in Special Needs (15 ECTS). *Main supervisor*

Carlbring, J. (2020) Department of Computer and Information Science, Linköping University. "Inclusive Design for Mobile Devices with WCAG and Attentional Resources in Mind". Master's Thesis in Cognitive Science (30 ECTS). *Main supervisor*

Ma, L. (2015). Department of Psychology, Umeå University. "Cross-Modal Matching of Distractibility in Auditory and Tactile Stimuli". Master's Thesis in Cognitive Science (15 ECTS). *Co-supervisor*

Blide, M. (2014). Department of Psychology, Umeå University. "Att orka lämna ett misshandelsförhållande: Anknytningens beydelse (To cope leaving abusive relationships: The importance of attachment). Master's Thesis in Clinical Psychology (30 ECTS). *Co-supervisor*

Teaching Workshops

Sep 2026	**Introduction to Python Programming**
	Disability Research Division, Department of Behavioural Sciences and Learning, Linköping University.
	An introduction to Python ogramming language - presented to researchers and PhD students with a focus on basic programming using Python.
Oct 2018	**R Workshop "Step-by-Step"**

Department of Psychology, Umeå University.

An introduction to R statistical programming language - presented to senior researchers with a focus on basic programming and the R environment.

Additional skills

- Extensive knowledge in statistical software such as SPSS, JASP, and R statistical programming environment
- Strong scripting skills in Python (v2.7.x & v3.x.x) and R
- Substantial skills in programming and performing experiments using both E-prime and Python (i.e., PsychoPy, OpenSesame, & Expyriment)
- Good skills in Microsoft Word and Excel
- Basic programming skills in Visual Basic, E-basic (E-prime), MATLAB, Bash, JavaScript, and PHP
- Basic skills in Markdown (e.g., RMarkdown) and LaTeX

Responsibilities

- Organizer of the monthly HEAD/DRD Seminar Series (e.g., contacting and inviting national and international researchers, booking tickets and accommodation)
- Elected chairperson of the Ph.D. students Council of the Department of Psychology, Umeå University.

Table 1: Teaching responsibilities - an overview of type of teaching, hours, etc.

Period	Subject	Type	Clock Hours	Course/Program	Level	Language
VT 2014	Scientific communication, research methods, research ethics	Lectures, seminars, supervision	60	Psychological test and research methods/Statistical and Empirial Methods, the Clinical Psychologist program/the Bachelor's Program in Cognitive Science	UG	Swe
	Cognitive Psychology	Supervision of projects	20	Introduction to Psychology, the Clincial Psychologist Program	UG	Swe
HT 2014	Clinical Psychology	Co-supervision of thesis project	20	Master Thesis in Psychology, 30 ECTS, the Clinical Psychologist Program	G	Eng
	Applied Cognitive Science	Supervision of projects	40	Project in Cognitive Science, the Bachelor's Program in Cognitive Science	UG	Swe
	Cognitive Psychology	Supervision of projects	20	Introduction to Psychology, the Clincial Psychologist Program	UG	Swe
VT 2015	Cognitive Psychology	Supervision of projects	24	Introduction to Psychology, the Clincial Psychologist Program	UG	Swe
	Cognitive Psychology	Lectures, seminars, supervision of projects	20	Basic psychology and sport psychology, the Coaching Program	UG	Swe
	Cognitive Psychology	Supervision of thesis project	15	Master Thesis in Cognitive Science, 15 ECTS, the Master's Program in Cognitive Science	G	Eng
	Applied Cognitive Science	Supervision of projects	110	Project in Cognitive Science, the Bachelor's Program in Cognitive Science	UG	Swe
HT 2015	Cognitive Psychology	Supervision of projects	24	Cognitive Psychology, the Clinical Psychologist Program	UG	Swe

Table 1: Teaching responsibilities - an overview of type of teaching, hours, etc. *(continued)*

Period	Subject	Type	Clock Hours	Course/Program	Level	Language
	Qualitative methods	Seminars	20	the Psychology of Organizations, Environment and Work, the Clinical Psychologist Program	UG	Swe
VT 2016	Cognitive Psychology	Lab demonstrations	30	2:1 Cognition, the Clinical Psychologist Program	UG	Eng
	Cognitive Psychology: Attention	Lectures, seminars, supervision of projects	60	Applied Cognitive Psychology, the Bachelor's Program in Cognitive Science	UG	Swe
	Applied Cognitive Science	Supervision of projects	110	Project in Cognitive Science, the Bachelor's Program in Cognitive Science	UG	Swe
	Qualitative and Quantitative methods	Seminars, computer labs	30	the Psychology of Organizations, Environment and Work, the Clinical Psychologist Program	UG	Swe
HT 2016	Cognitive Psychology	Lab demonstrations	30	2:1 Cognition, the Clinical Psychologist Program	UG	Eng
	Qualitative and Quantitative methods	Seminars, computer labs	30	the Psychology of Organizations, Environment and Work, the Clinical Psychologist Program	UG	Swe
	Cognitive Psychology: Perception, Attention, and Consciousness	Lectures	24	Perception, the Bachelor's Program in Cognitive Science	UG	Swe
	Clinical Psychology	Seminars	9	Master Thesis in Psychology, 30 ECTS, the Clinical Psychologist Program	G	Swe

Table 1: Teaching responsibilities - an overview of type of teaching, hours, etc. *(continued)*

Period	Subject	Type	Clock Hours	Course/Program	Level	Language
HT 2017	Cognitive Psychology	Lab demonstrations	30	2:1 Cognition, the Clinical Psychologist Program	UG	Eng
VT 2017	Cognitive Psychology, Applied: Attention	Lectures, seminar	20	Applied Cognitive Psychology, the Bachelor's Program in Cognitive Science	UG	Swe
	Applied Cognitive Science	Supervision of projects	40	Project in Cognitive Science, the Bachelor's Program in Cognitive Science	UG	Swe
	Qualatitive and Quantative research methods	Seminars, computer labs	30	the Psychology of Organizations, Environment and Work, the Clinical Psychologist Program	UG	Swe
	Clinical Psychology	Seminars	30	Master Thesis in Psychology, 30 ECTS, the Clinical Psychologist Program	G	Swe
	Cognitive Psychology	Lab demonstrations	30	2:1 Cognition, the Clinical Psychologist Program	UG	Eng
	Cognitive Psychology: Attention and Consciousness	Lectures, seminars	24	Perception, the Bachelor's Program in Cognitive Science	UG	Swe
	Developmental Psychology	Seminar leader	50	Learning and teaching, pedagogy	UG	Swe
VT 2019	Research Methods	Workshops in SPSS	18	Human Resources, Special Pedagogists	G	Swe
	Applied Cognitive Science	Supervision of Projects	24	Applied Cognitive Science, the Bachelor's Program in Cognitive Science	UG	Eng
	Pedagogy	Examination of Thesis	6	Masther Thesis in Special Pedagogy, 30 ECTS, the Specialist Pedagogy Program	G	Swe

Table 1: Teaching responsibilities - an overview of type of teaching, hours, etc. *(continued)*

Period	Subject	Type	Clock Hours	Course/Program	Level	Language
HT 2019	Quantative methods, theory of science	Seminar leader, grading exams	62	Theory of Science and Research Methods, Pedagogy	UG	Swe
	Cognitive Science	Seminar leader, grading exams	20	Cognitive Science - Methods, the Master's Program in Cognitive Science	G	Swe
	Disability Research	Lectures, workshops, examining	40	Theory of Science and Research Methods, Disability Research II and III	UG	Swe
	Perception	Lectures	12	Neuropsychology and Neuroscience, the Clinical Psychologist Program	G	Swe
	Applied Learning Disabilities	Supervision of groups (Problem-based learning)	16	Cognitive Psychology, the Clinical Psychologists Program	UG	Swe
	Adult life and Aging	Supervision of groups (Problem-based learning)	20	Developmental Psychology, the Clinical Psychologists Program	UG	Swe
HT and VT 2020	Quantative methods	Lectures	92	Methods for Research and Development in Educational Institutions, Pedagogy	G	Swe
VT 2020	Applied Cognitive Science	Supervision	24	Applied Cognitive Science, the Bachelor's Program in Cognitive Science	UG	Eng
	Cognitive Science	Supervision, Examinating	45	Bachelor's Thesis in Cognitive Science, the Bachelor's Program in Cognitive Science	UG	Swe
	Cognitive Science	Supervision	20	Master's Thesis in Cognitive Science, the Master's Program in Cognitive Science	G	Swe

Table 1: Teaching responsibilities - an overview of type of teaching, hours, etc. *(continued)*

Period	Subject	Type	Clock Hours	Course/Program	Level	Language
HT 2020	Disability Research	Lectures, workshops, examining	80	Theory of Science and Research Methods, Disability Research II and III	UG	Swe
	Quantative methods, theory of science	Seminar leader, grading exams	50	Theory of Science and Research Methods, Pedagogy	UG	Swe
VT 2021	Applied Cognitive Science	Supervision	20	Applied Cognitive Science, the Bachelor's Program in Cognitive Science	UG	Eng
		Supervision of research project	8	Research Project, the Master's Programme in Cognitive Science	G	Swe
	Cognitive Science	Supervision, Examinating	45	Bachelor's Thesis in Cognitive Science, the Bachelor's Program in Cognitive Science	UG	Swe
	Disability Research	Supervision of theses	30	University Diploma Thesis, Disability Research II	UG	Swe
HT 2021	Disability Research	Lectures, workshops, examining	80	Theory of Science and Research Methods, Disability Research II and III	UG	Swe
	Disability Research	Lectures, examination	40	Introduction to Research methods and Ethics	UG	Swe
	Research Methods	Supervision of groups (Problem-based learning)	36	Research Methods, the Clinical Psychologists Program	UG	Swe
	Applied Learning Disabilities	Supervision of groups (Problem-based learning)	16	Cognitive Psychology, the Clinical Psychologists Program	UG	Swe
	Adult life and Aging	Supervision of groups (Problem-based learning)	20	Developmental Psychology, the Clinical Psychologists Program	UG	Swe
VT 2022	Applied Cognitive Science	Supervision	20	Applied Cognitive Science, the Bachelor's Program in Cognitive Science	UG	Eng

Table 1: Teaching responsibilities - an overview of type of teaching, hours, etc. *(continued)*

Period	Subject	Type	Clock Hours	Course/Program	Level	Language
VT 2023	Cognitive Science	Course leader, Lectures, Seminars, Supervision, Examinator	80	Evaluation and Cognitive Measurement	G	Swe
	Cognitive Science	Course leader, Lectures, Seminars, Supervision, Examinator	80	Advanced Statistics	G	Swe
	Cognitive Science	Course leader, Seminar Leader, Supervision, Examinator	68	Cognitive Science Project I	G	Swe
	Disability Research	Supervision	46	Disability Research II	UG	Swe
	Cognitive Science	Supervision, Examinating	45	Bachelor's Thesis in Cognitive Science, the Bachelor's Program in Cognitive Science	UG	Swe
HT 2023	Cognitive Science	Course Leader, Lectures, Seminars, Examinator	231	Quasi-Experimentation and Statistical Analysis	UG	Eng
	Cognitive Science	Seminars, Group supervisions	36	Cognitive Science - Methods	G	Swe
	Clinical Psychology	Supervision of groups (Problem-based learning)	16	Cognitive Psychology 2 - Neuropsychology and Cognitive Neuroscience	UG	Swe
	Clinical Psychology	Supervision of groups (Problem-based learning)	16	Developmental Psychology 2 - Infants and preschool age	UG	Swe
	Clinical Psychology	Supervision of groups (Problem-based learning)	16	Organizational psychology 1 - foundations and theories	UG	Swe
	Clinical Psychology	Supervision of groups (Problem-based learning)	16	Group and Social Psychology – Applied knowledge	UG	Swe
	Clinical Psychology	Run R-scripts, read reports	20	Research Methods	UG	Swe
VT 2024	Disability Research	Supervision of theses	36	Disability Research II	UG	Swe
	Cognitive Science	Course leader, Seminar Leader, Supervision, Examinator	45	Cognitive Science Project I	G	Swe

Table 1: Teaching responsibilities - an overview of type of teaching, hours, etc. *(continued)*

Period	Subject	Type	Clock Hours	Course/Program	Level	Language
	Cognitive Science	Course Leader, Lectures, Seminars, Examinator	30	Evaluation and Cognitive Measurement	G	Swe
	Applied Cognitive Science	Supervision	42	Applied Cognitive Science, the Bachelor's Program in Cognitive Science	UG	Eng
	Cognitive Science	Supervision, Examinating	47	Bachelor's Thesis in Cognitive Science, the Bachelor's Program in Cognitive Science	UG	Swe
	Human Resources Program	Lecture, seminars, labs	56	Research Methodology	UG	Swe
	Special Education	Supervision of thesis project	24	Master's Thesis in Special Needs Education	G	Swe
HT 2024	Cognitive Science	Course Leader, Lectures, Examinator	83	Quasi-Experimentation and Statistical Analysis	UG	Eng
	Cognitive Science	Seminars, Group supervisions	36	Cognitive Science - Methods	G	Swe
	Clinical Psychology	Supervision of groups (Problem-based learning)	16	Cognitive Psychology 2 - Neuropsychology and Cognitive Neuroscience	UG	Swe
	Disability Research	Lectures	33	Disability Research I	UG	Swe
VT 2025	Cognitive Science	Course leader, Lectures, Seminars, Supervision, Examinator	70	Advanced Statistics	G	Swe
	Cognitive Science	Course leader, Seminar Leader, Supervision, Examinator	132	Cognitive Science Project I	G	Swe
	Disability Research	Supervision	62	Disability Research II	UG	Swe
	Cognitive Science	Course Leader, Lectures, Seminars, Examinator	57	Evaluation and Cognitive Measurement	G	Swe
	Applied Cognitive Science	Supervision	21	Applied Cognitive Science, the Bachelor's Program in Cognitive Science	UG	Eng

Table 1: Teaching responsibilities - an overview of type of teaching, hours, etc. *(continued)*

Period	Subject	Type	Clock Hours	Course/Program	Level	Language
	Disability Research	Lectures	46	Disability Research I	UG	Swe
	Special Needs Education	Supervision of thesis project	24	Master's Thesis in Special Needs Education	G	Swe
	Human Resources Program	Lecture, seminars, labs	86	Research Methodology	UG	Swe
	Disability Research	Supervision of thesis project	30	Bachelor's Thesis in Disability Research	G	Swe

Note:
Total of clock hours: 3440 , UG = Undergraduate, G = Graduate, VT = Spring term, HT = autumn term, * = two courses.